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ICT and economic growth – Comparing developing, emerging and developed countries

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ABSTRACT

This paper analyzes the impact of information and communication technologies (ICT) on economic growth in developing, emerging and developed countries. The main question is whether the gains from investments in ICT differ between developing, emerging and developed countries. The analysis is based on a high-quality sample of 59 countries for the period 1995–2010. Various panel data regressions confirm the previously reported positive relationship between ICT capital and GDP growth. For the combined sample of all 59 countries, the estimated output elasticity of ICT is larger than the ICT factor compensation share suggesting excess returns to ICT capital. The regressions for the subsamples of developing, emerging and developed countries do not reveal statistically significant differences in the output elasticity of ICT between these three groups of countries. Thus, the results indicate that developing and emerging countries are not gaining more from investments in ICT than developed economies, calling into question the argument that these countries are 'leapfrogging' through ICT.

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1. Introduction

Productivity growth lays the foundation for improvements in the standard of living.[2] Investments in information and communication technologies (ICT) are seen as a key driver of productivity growth. This relationship has been extensively studied in developed countries at the firm, industry and country level with the majority of studies showing the productivity effect of ICT to be positive and economically significant.[3] Recent literature reviews by Cardona et al. (2013), Draca, Sadun, and Van Reenen (2007), and Van Reenen, Bloom, Draca, Kretschmer, and Sadun (2010) list a comprehensive set of studies applying different methodologies. To date, there has been only rather weak and ambiguous empirical evidence on the contribution of ICT investments to economic growth for emerging and, in particular, developing countries. Despite the rather ambiguous empirical evidence, the World Bank (2012) takes an optimistic view stating that "information and communication technologies (ICTs) have great promise to reduce poverty, increase productivity, boost economic growth (...)". The weak and ambiguous empirical evidence on the impact of ICT in developing countries may largely be driven by the lack of high-quality micro- and macro-level data sets on ICT for these countries.

A priori, there may be valid reasons why the impact of ICT on growth in developing and emerging countries is different than in developed countries. On the one hand, developing and emerging countries might be lacking absorptive capacities such as an appropriate level of human capital or other complementarity factors such as R&D expenditures[4] and therefore gain less than developed countries from investments in ICT. On the other hand, ICT could enable developing and emerging countries to 'leapfrog' traditional methods of increasing productivity as mentioned by Steinmueller (2001). The additional productivity gains could be triggered by "ICT-related spillovers or network effects"[5] as ICT may lower transaction costs and speed up the process of knowledge creation.[6] These network effects may be more pronounced "when many firms in a region or industry are using similar levels or types of ICT".[7]

This paper contributes to the literature in several ways. Section 2 reviews the current empirical literature on the impact of ICT on economic growth, focusing on differences in methodologies, data sources and sample periods. Section 3 describes the unique features of the data set used for the empirical work. The Confer-

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[2] See e.g. Timmer, Inklaar, O'Mahony, and Van Ark (2010), p. 18.

[3] Cardona, Kretschmer, and Strobel (2013), p. 109.

[4] Keller (2004), p. 774.

[5] Stiroh (2002), p. 43.

[6] Pilat (2004), pp. 57–58.

[7] Draca et al. (2007), p. 106.

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ence Board Total Economy Database (The Conference Board, 2014a), the main source, contains annual data for GDP, ICT and non-ICT capital services as well as labor services to control for changes in human capital. In contrast to previous studies, the empirical work is based on these capital and labor services which are a more appropriate measure than stock variables. The sample for the empirical analysis consists of 59 developing, emerging and developed countries between the years of 1995 and 2010 and thus covers more countries and years than previous studies were able to. This rich, high-quality data set allows for a clarification of the previous empirical results. Based on this data, Section 4 presents the results, including a comparison of the estimated coefficients for the output elasticity of ICT for the pooled sample and the three subgroups of countries.

The results for the full sample of countries confirm the positive contribution of ICT to economic growth with an output elasticity of about 10%, exceeding the factor compensation share by a considerable amount.[8] The estimates with subsamples for the three country groups only reveal small differences between developing, emerging and developed countries. Thus, the results indicate that developing and emerging countries are not gaining more from investments in ICT than developed economies, calling into question the 'leapfrogging' through ICT argument discussed above.

2. Related literature

The macro-level empirical work on the relationship between ICT and economic growth is mostly based on growth accounting as well as econometric studies. Several growth accounting studies reveal economically significant contributions of ICT capital to economic growth after the mid-1990s in developed economies. These studies by Jorgenson and Stiroh (2000) and Oliner and Sichel (2000, 2002) focused on the productivity effects of ICT in the US because "European statistics offices' published industry data on ICT assets lag behind the US".[9] The work by Inklaar, O'Mahony, and Timmer (2005) compared the ICT contribution of the US and the EU4 consisting of France, Germany, the Netherlands and the United Kingdom, showing higher contributions for the US than the EU4 during the period 1979–2000. With the release of the EU KLEMS database (O'Mahony & Timmer, 2009), cross-country studies like Inklaar, Timmer, and Van Ark (2008), Van Ark, O'Mahony, and Timmer (2008), and Strauss and Samkharadze (2011) as well as Timmer, Inklaar, O'Mahony, and Van Ark (2011) appeared, showing substantial sectoral and cross-country heterogeneity with respect to the contribution of ICT to labor productivity growth in developed countries.

In the past decade, a number of macro-level econometric studies on ICT and productivity in developed countries have been carried out. Stiroh (2002) surprisingly finds a negative output elasticity of ICT capital in his pooled OLS and IV regressions based on US manufacturing industries data for the years 1984–1999. With an updated data set and more detailed industry breakdown, Stiroh (2005) reports positive coefficients of ICT capital in the production function regressions. Based on dynamic panel data estimations, O'Mahony and Vecchi (2005) as well as Dimelis and Papaioannou (2011) show that there is a significant positive effect of ICT capital on output growth for both the UK and the US. Dahl, Kongsted, and Srensen (2011) confirm these findings for eight European countries using EU KLEMS data (O'Mahony & Timmer, 2009).

Another strand of the literature focuses solely on communication technologies (CT). Roller and Waverman (2001) find a causal relationship between wireline telecommunications and GDP for 21 OECD countries. Czernich, Falck, Kretschmer, and Woessmann (2011) support the finding of Roller and Waverman (2001) on the importance of communication technologies. Based on a panel of 20 OECD countries, they provide empirical evidence that an increase in the level of broadband penetration leads to an increase in GDP growth rates. A recent literature survey conducted by Bertschek, Briglauer, Hüschelrath, Kauf, and Niebel (2015) confirms the positive relationship between broadband internet and economic growth. More comprehensive reviews of the empirical literature on general ICT with a focus on developed countries are found in Biagi (2013), Cardona et al. (2013), Draca et al. (2007), and Van Reenen et al. (2010). Furthermore, Indjikian and Siegel (2005) provide an overview of the quantitative and qualitative research on the relationship between ICT and economic performance in both developed and developing countries.

A priori, it is not clear whether the impact of ICT on economic growth in emerging and developing countries is larger than in developed countries. Steinmueller (2001), p. 194 points out that "ICTs have the potential to support the development strategy of 'leapfrogging', i.e. bypassing some of the processes of accumulation of human capabilities and fixed investment in order to narrow the gaps in productivity and output that separate industrialized and developing countries." If the 'leapfrogging' hypothesis holds, the output elasticities of ICT in developing and emerging countries should be significantly larger than those in developed countries. Whether this strategy is successful crucially depends on the absorptive capacities ("the ability and effort of workers and managers to apply new technology"[10]) of the emerging and developing countries (see e.g. Keller, 1996; Keller, 2004 & Henry et al., Henry, Kneller, & Milner, 2009). A report by the United Nations (2011), pp. 71–78 lists a variety of examples illustrating why ICT may have a strong(er) impact on economic performance in emerging and developing countries. First of all, investments in ICT may decrease the administrative burden on firms through the introduction of e-government applications. Furthermore, ICT can be used for training and advisory services. It also enhances access to relevant information. As a tool to save travel time and to reduce transaction costs, mobile money services are particularly important to micro- and small enterprises. None of these ICT services and applications are specific to emerging and developing countries. But in these countries, ICT often provides services that were previously not available in either the digital or non-digital economy.

The empirical macro-level literature on ICT and growth in developing and emerging countries examines differing country groups and time periods, which limits the comparability and generalizability of the results. Dewan and Kraemer (2000) find a positive effect of the ICT capital stock on GDP growth in developed countries, whereas the ICT coefficient for developing countries is insignificant. The authors explain this finding through potentially missing 'IT-enhancing complementarity factors' like human capital. The estimation is performed with a panel of 36 countries (14 developing and 22 developed countries) for the years 1985–1993. As this period represents just the beginning of the rapid diffusion of ICT in developed countries, it was probably too early to see any (economically) significant effects in developing countries. Pohjola (2002), with data on 42 countries within the period 1985–1999, does not find any significant relationship between ICT and economic growth in either the full sample or any of the country subgroups. These results are possibly driven by the fact that the ICT input variable is measured as the share of nominal ICT investment in GDP, which does not incorporate any quality improvements of ICT over time. Papaioannou and Dimelis (2007)

[8] This difference is smaller in the IV regressions.

[9] Draca et al. (2007), p. 112.

[10] Kneller (2005).

show that the impact of the ICT capital stock on labor productivity growth is stronger in developed than in developing countries. Their analysis is based on an Arellano and Bond (1991) panel data estimator applied to a sample of 22 developed and 20 developing countries for the period 1993–2001. Based on the same data but with a refined econometric approach and the inclusion of foreign direct investment (FDI) as an additional control variable, since it is seen as an important channel for technology diffusion,[11] Dimelis and Papaioannou (2010) report that the impact of ICT is stronger in developing than in developed countries. The study by Yousefi (2011) uses World Bank data for the period 2000–2006 and finds an insignificant impact of ICT capital investment on output growth for developing countries. The paper by Dedrick, Kraemer, and Shih (2013) has the most recent and comprehensive data so far. Their data set consists of 45 upper-income developing and developed countries for the period 1994–2007. Their set of so-called 'upper-income developing countries' is comparable to the emerging countries in the present study. They provide econometric evidence for the contribution of ICT to growth for both developing and developed countries with slightly larger output elasticities of ICT stock in developed than in the upper-income developing countries. As mentioned before, the rather ambiguous empirical evidence so far might be explained by different analytical approaches and the use of data sets covering different countries and time periods.[12]

Insignificant output elasticities of ICT capital, especially in developing and emerging countries, could be caused by a lack of absorptive capacities or the fact that it may take some time for the gains from ICT to materialize. Additional productivity gains, with ICT output elasticities higher than the corresponding factor shares, could be triggered by spillovers, network effects or other positive externalities. If the 'leapfrogging' hypothesis holds, the output elasticities of ICT in developing and emerging countries should be significantly larger than those in developed countries.

Apart from the econometric analyses, several growth accounting studies by Jorgenson and Vu (2005, 2010, 2011, 2016) focus on specific regional groups of countries like 'Developing Asia' and Latin America and therefore implicitly perform a comparison of the contribution of ICT to growth in developing, emerging and developed countries. The 2016 World Development Report (World Bank, 2016, p. 13) presents results of a growth accounting exercise for developed and developing countries for four time periods between 1995 and 2014. It shows that the share of ICT capital in the total input factor contribution to GDP growth is very similar in developing and developed countries.

Furthermore, a number of micro-level studies analyze the impact of CT/ICT on productivity for certain developing and emerging countries. Aker and Mbiti (2010) thoroughly discuss, based on the fact that mobile phone usage in Sub-Saharan Africa has grown significantly over the past decade, the (potential) economic implications of this rapid diffusion. Several other studies examine the economic impact of mobile phones especially in the context of small and micro-enterprises in developing and emerging countries. Jensen (2007), based on data from India, finds welfare increases through mobile phones for both consumers and producers. Muto and Yamano (2009) find that improved mobile phone coverage in remote areas of Uganda increased market participation of farmers that produce perishable goods. On the other hand, Tadesse and Bahiigwa (2015) find almost no significant effects of mobile phone access on the marketing decisions of Ethiopian farmers. Furthermore, there are some studies which use more general ICT indicators and also focus on manufacturing and services firms. Commander, Harrison, and Menezes-Filho (2011) find a positive relationship between ICT capital and the productivity of firms in Brazil and India. Based on data for 117 developing and emerging countries from the World Bank Enterprise Surveys, Paunov and Rollo (2016) show that there is a clear positive impact of industries' internet use on firm labor productivity. A recent study by Cirera, Lage, and Sabetti (2016) finds a positive impact of ICT on innovation, but ambiguous results for the link between innovation and productivity, for a sample of six African countries.

3. Data and methodology

The primary data source for the analysis is the Conference Board Total Economy Database[13] (The Conference Board, 2014a). It was originally developed by the Groningen Growth and Development Centre[14] and has been maintained by the Conference Board since 2007. The database contains annual data for i.a. GDP, ICT and non-ICT capital services as well as labor services for 123 countries. Output data are available for the period 1950–2013, whereas capital input and detailed labor input data are generally only available for a subset of countries for the years 1990–2013. The final data set for the empirical analysis consists of 59 countries for the period 1995–2010 with a total of 893 observations. There are two reasons for restricting the sample to the period 1995–2010. First, years prior to 1995 often have missing values for the ICT input variables, especially in developing countries. Second, capital input data in the Total Economy Database after 2010 are often inferred from different sources or calculated with a slightly different methodology, which could reduce the overall data quality for the most recent years. Further control variables for the robustness checks are taken from the World Bank World Development Indicators (WDI).[15]

3.1. Categorization of countries

As the primary goal of this empirical work is to compare the contribution of ICT to economic growth for developing, emerging and developed countries, it is necessary to divide the total sample into three reasonable subsamples. The definition of the country groups is usually based on GDP/GNI per capita or more general indicators like the literacy rate of the countries. The threshold variable chosen in this empirical application is GDP per capita in the starting year of the data set 1995 expressed in purchasing power parity (PPP) adjusted US dollars from 2013. As shown in Table 3.1, there are 18 developing countries, 22 emerging and 19 developed countries. According to this definition, countries with less than 6500 2013 US dollars in GDP per capita are classified as developing countries, and all countries with more than 23,000 2013 US dollars in GDP per capita are classified as developed countries. There is quite a large gap in GDP per capita between Peru, as the developing country with the largest GDP per capita, and South Africa as the emerging country with the smallest GDP per capita. The same is true for the split between emerging and developed countries. Portugal as the emerging country in 1995 with the largest GDP per capita and Ireland as the developed country with the smallest GDP per capita differ in this value by more than 20%. A threshold test as described in Hansen (2000) confirms the threshold between developing and emerging countries.[16] The GDP per capita values of

[11] See e.g. Keller (2004), p. 752.

[12] In contrast to the one-time compilation of data sets used in previous studies, the present empirical work is based on an annually updated publicly available data set on ICT and economic growth. See Section 3 for details.

[13] To be precise, I use the March 2014 update of the Conference Board Total Economy Database January 2014 release. See https://www.conference-board.org/-data/economydatabase/index.cfm?id=30565. I would like to thank Klaas De Vries for providing this update covering the majority of errors I found in the January 2014 release.

[14] See http://www.rug.nl/ggdc/overview-databases/ted.

[15] World Bank (2014).

[16] The test was conducted by the Stata package thresholdtest.ado available at http://www.ssc.wisc.edu/~bhansen/progs/progs_threshold.html – version March 24, 2014.

Table 3.1
List of countries by group: developed >23,000 of 2013 US$ GDP per Capita, developing <6500 of 2013 US$ GDP per Capita in year 1995.

Developing				Emerging				Developed			
1.	Bangladesh	BGD	998	1.	Argentina	ARG	10,442	1.	Australia	AUS	33,075
2.	Bolivia	BOL	3529	2.	Brazil	BRA	7770	2.	Austria	AUT	33,166
3.	Cameroon	CMR	1740	3.	Bulgaria	BGR	7244	3.	Belgium	BEL	31,928
4.	China	CHN	2646	4.	Chile	CHL	10,110	4.	Canada	CAN	33,055
5.	Egypt	EGY	3611	5.	Colombia	COL	7118	5.	Denmark	DNK	33,922
6.	India	IND	1704	6.	Costa Rica	CRI	9013	6.	Finland	FIN	26,062
7.	Indonesia	IDN	3510	7.	Czech Republic	CZE	17,647	7.	France	FRA	30,257
8.	Jordan	JOR	3998	8.	Ecuador	ECU	6733	8.	Germany	DEU	32,232
9.	Kenya	KEN	1297	9.	Hungary	HUN	13,210	9.	Ireland	IRL	23,913
10.	Morocco	MAR	2534	10.	Iran	IRN	7742	10.	Italy	ITA	30,389
11.	Nigeria	NGA	1123	11.	Jamaica	JAM	10,734	11.	Japan	JPN	33,270
12.	Pakistan	PAK	2034	12.	Malaysia	MYS	10,534	12.	Netherlands	NLD	33,811
13.	Peru	PER	5570	13.	Mexico	MEX	11,025	13.	New Zealand	NZL	25,146
14.	Philippines	PHL	2768	14.	Poland	POL	10,255	14.	Norway	NOR	44,618
15.	Senegal	SEN	1264	15.	Portugal	PRT	19,041	15.	Spain	ESP	24,720
16.	Sri Lanka	LKA	2698	16.	Slovak Republic	SVK	12,203	16.	Sweden	SWE	30,034
17.	Tunisia	TUN	4738	17.	Slovenia	SVN	18,234	17.	Switzerland	CHE	39,646
18.	Vietnam	VNM	1655	18.	South Africa	ZAF	6521	18.	United Kingdom	GBR	29,972
				19.	South Korea	KOR	17,429	19.	United States	USA	40,427
				20.	Thailand	THA	7118				
				21.	Turkey	TUR	8484				
				22.	Venezuela	VEN	9970				

each country at the end of the sample period (2010) are reported in Table A.1 in the Appendix.

3.2. Outlier detection

There are multiple reasons for outlier problems within such a rich cross-country data set. One could be a break in the raw input data series caused by switching from one data source to another. This will clearly affect the growth rate of e.g. labor input in this particular year. Outliers could also occur when certain data points for certain countries are estimated or extrapolated. Another problem might be general measurement issues, especially in developing countries, due to a lack of resources in the national statistical offices. E.g. Klasen and Blades (2013) argue that "the measurement of economic and social performance has not kept pace with the apparent drastic improvements in that performance in recent years."

To account for breaks in the data series and more general data errors, I employ a five step approach. First, I drop countries with erroneous or implausible data during the whole period. This reduces the number of countries with data on ICT input from 68 to 59 countries.[17] The next step is to drop the single observations (years) of the remaining 59 countries with obvious data errors such as a zero ICT capital compensation share (12 observations). Visual inspection of year by year scatter plots between the GDP growth rate and the growth rates of factor input variables revealed 11 additional problematic observations. The fourth step is to calculate the so-called DFBETA[18] values. The idea behind this method is to calculate the impact of the *i*th observation on the regression coefficient and drop the observations with abnormally high impact (9 observations dropped). The methodology proposed by Hadi (1992, 1994) is another commonly used approach to detect outliers in the empirical literature.[19] This fifth step detects another 12 observations as outliers. The complete list of the 44 dropped observations and the approach used to identify each outlier is available in Table A.2. Moreover, Jamaica and Iran have a total of 7 missing observations for the variable 'Exports % GDP'. This gives a total of 893 instead of 944 observations (16 years and 59 countries).

3.3. Capital and labor input variables

In contrast to previous studies, the empirical work in this paper is based on capital services instead of capital stocks. Capital services are a more appropriate measure than capital stocks especially because of the diffusion of short-lived ICT assets. This is, e.g., supported by Inklaar and Timmer (2013), p. 13: "A capital services measure would reflect that shorter lived assets have a larger return in production, as indicated by the user cost of capital of each asset". At the level of a single asset, the flow of capital services and capital stocks are assumed to be strictly proportional. For aggregate measures of ICT and non-ICT capital, only capital services measures take the heterogeneity of assets into account (Schreyer, 2001, p. 47). The growth rates of ICT capital services as well as non-ICT capital services are calculated as the growth rates of the real net stocks of the single assets (information technology equipment, communication technology equipment and software for ICT) weighted by their factor shares in total ICT (non-ICT) capital compensation.

The labor input variable is the growth rate of labor services. It is the sum of the growth rate of the labor composition index and the growth rate of labor quantity. The growth rate of the labor composition reflects changes in human capital. It is constructed[20] as the growth rate of the share of different skill-level groupings in the labor force weighted by their share in total labor compensation. Labor services therefore account for differences in the marginal product of workers of different skill levels. The definition of the growth rate of labor quantity differs from country to country. In the more advanced economies, it is the growth rate of total hours worked. In contrast, the labor quantity for less developed countries is usually based on the employment growth rate. These two methodologies do not lead to any differences as long as the average hours worked per person do not change over time.[21]

Nearly all developing and some emerging countries are lacking data for labor compensation. The ad hoc approach of the The

[17] The countries not in the sample are Algeria (implausible labor services growth rates), Greece (overall data quality issues), Hong Kong and Singapore (very specific export-oriented type of economy), Israel, Romania and the Russian Federation (implausible non-ICT capital services growth rates), Taiwan (no data on export share) as well as Uruguay (implausible ICT capital services growth rates).

[18] See e.g. Temple (2000).

[19] See e.g. Ardizzi, Petraglia, Piacenza, and Turati (2014), Durham (2004), Harbaugh, Levinson, and Wilson (2002).

[20] See The Conference Board (2014b), p. 5.

[21] See The Conference Board (2014b), p. 2.

Conference Board (2014a) is to assume a labor compensation share in total factor compensation of 0.5. This assumption is justified by the fact that in economies where capital is scarcer, the returns for capital should be higher and the returns for labor lower.[22] As the labor compensation is not directly included in the regression, this does not influence the estimated output elasticities of ICT- and non-ICT capital. However, the potentially underestimated labor compensation share in developing countries might affect the comparison between the output elasticity of ICT and the growth accounting based ICT compensation share.

3.4. Descriptive statistics

Table 3.2 reports descriptive statistics of the three country subgroups for the sample period 1995–2010.[23] The average gross domestic product (GDP) per capita[24] in developing countries is 3325 US dollars with the minimum value just below 1000 US dollars (the value for Bangladesh in 1995) and the maximum for Peru in 2010 with 9009 US dollars. For the emerging countries, this average is 13,367 US dollars ranging from 6433 US dollars to 31,451 US dollars. As the GDP per capita value of 1995 defines the three country subgroups, we can see the highest average GDP per capita with 38,044 US dollars for the developed countries. The average growth rate of GDP is 5.1% in developing, 3.6% in emerging and 2.4% in developed countries indicating at least some kind of catch-up effect.

ICT capital services as the main variable of interest shows a similar average growth rate of 18 and 17% in developing and emerging countries respectively. This number is much lower in the group of developed economies, where the average growth rate is just 11%. The average growth rates of non-ICT capital services are much lower than those of ICT capital services, with the highest value of 4.8% found in developing countries and the lowest value of 2.3% seen again in the developed economies. The labor services growth rates range from 1.3% in developed countries to 2.8% in developing countries. This rate is defined as the sum of the growth rate of labor composition and labor quantity. The largest average growth rate of labor composition (i.e. the accumulation of human capital) occurs in emerging countries, followed by developed countries. The labor composition growth rate in developing countries is only half as high as the growth rate in emerging countries. Hence, the high labor services growth rates in developing countries are mainly a result of the growing labor quantity and only to a lesser extent driven by improvements in the average skill-levels of employees.

The average factor compensation share of ICT capital[25] ranges in all three country groups between 4 and 5%, with the highest value in the developed economies.[26] This is clearly an interesting result on its own. The within-group variation in developing and emerging countries is much larger than in developed countries. The variation in the average ICT compensation share in total factor compensation over time is displayed in Fig. 3.1, showing a slight upward trend for the emerging countries and, in contrast, a minimal downward trend for the developed countries.[27]

Fig. 3.2 shows a scatterplot of the relationship between the average GDP growth rates and the average ICT capital services growth rates between 1995 and 2010 for the three country subgroups and the full sample. All country groups display a positive relationship between GDP and ICT capital services growth rates, with developing countries having the steepest slope for the fitted values.

3.5. Econometric model

As shown in the descriptive statistics, the average factor compensation shares of ICT capital in total factor compensation are nearly identical in developing, emerging and developed countries, with values ranging from 0.04 to 0.05. Regression-based output elasticities are able to reveal whether there are excess returns of investments in ICT capital and whether these returns differ between the country subgroups. For the comparison of output elasticities between the three country groups, following e.g. Stiroh (2002), an (augmented) Cobb-Douglas production function without imposing constant returns to scale is estimated as follows:

$$\Delta \ln Y_{c,t} = \beta_{ICT} \Delta \ln K^{ICT}_{c,t} + \beta_{NICT} \Delta \ln K^{NICT}_{c,t} + \beta_L \Delta \ln L_{c,t} + \beta_X \mathbf{X_{c,t}} + \lambda_t + \mu_c + \epsilon_{c,t} \quad (1)$$

with $\Delta \ln Y_{c,t}$ being the growth rate of GDP and $\Delta \ln K^{ICT}_{c,t}, \Delta \ln K^{NICT}_{c,t}, \Delta \ln L_{c,t}$ the growth rates of the input factors ICT capital services, non-ICT capital services and labor services in country c at time t. X denotes additional control variables in an augmented Cobb-Douglas production function setting controlling for differences in the production technology between countries. λ_t are time dummies which also capture possible non-linear time trends or global shocks, whereas μ_c indicates country dummies (in the fixed effects setting). $\epsilon_{c,t}$ is the general error term. The time dummies control for common shocks affecting (almost) all countries such as the global financial crisis of 2007–2008. Notwithstanding certain drawbacks, such as the substitution elasticity between inputs equaling one, the Cobb–Douglas function is used for most empirical estimations in the context of ICT and productivity.[28] For this particular research question, the elasticity of substitution between input factors is less relevant. Therefore, a Cobb–Douglas production function seems to be favorable over more flexible but also more complex functional forms like CES or translog production functions.

The econometric analysis is carried out for the full sample as well as for the subsamples of country groups. If the 'leapfrogging' effect described by Steinmueller (2001) holds, the output elasticities of ICT in developing and emerging countries should be significantly larger than those in developed countries. For the full sample regressions, I use four different estimators to estimate the Cobb-Douglas production function. The baseline specification is a pooled OLS (POLS) regression. In addition, I use a random effects (RE) estimator as well as a fixed effects (FE) panel regression model, with the latter controlling for unmeasured cross-country differences. The fourth specification features a panel IV approach with lagged growth rates of ICT capital services as instruments. This controls, albeit not as perfectly as an external instrument, for the endogeneity of the ICT input variables. Estimating (fixed effects) panel IV regressions are fairly data-demanding. The split-sample regressions substantially reduce the number of observations, leading to mostly insignificant coefficients. Therefore, the IV approach is not feasible for the regressions of the country subgroups.

[22] See The Conference Board (2014b) page 12. See also Tables A.3, A.4 and A.5.

[23] Detailed descriptive statistics are shown in Tables A.6, A.3, A.4 and A.5 in the Appendix.

[24] Expressed in purchasing power parity (PPP) adjusted US dollars from 2013.

[25] The country with the extraordinarily low ICT capital compensation share is Iran in 1995.

[26] The two period averages of ICT and non-ICT capital compensation shares shown in Tables A.3,A.4 are not directly available from the Total Economy Database and therefore calculated recursively: two period average ICT capital compensation = GDP contribution of ICT/$\Delta \ln$ ICT capital services.

[27] The non-ICT compensation share is very stable over time for all country subgroups as shown in A.1 in the Appendix.

[28] See e.g. the meta-survey by Cardona et al. (2013), p.112.

Table 3.2
Summary statistics: country groups – 1995–2010.

	Developing		Emerging		Developed		Total	
	N	Mean	N	Mean	N	Mean	N	Mean
GDP per Capita	280	3325	316	13,367	297	38,044	893	18,426
Δ ln(GDP)	280	5.1	316	3.6	297	2.4	893	3.7
Δ ln(ICT Cap. Serv)	280	18	316	17	297	11	893	15
Δ ln(N.ICT Cap. Serv.)	280	4.8	316	3.9	297	2.3	893	3.6
Δ ln(Labor Serv.)	280	2.8	316	1.9	297	1.3	893	2
Δ ln(Labor Composition)	280	0.25	316	0.48	297	0.4	893	0.38
Δ ln(Labor Quantity)	280	2.6	316	1.4	297	0.87	893	1.6
Δ ln(Employees)	280	2.5	316	1.5	297	1.2	893	1.7
Δ ln(Hours Worked)	46	2.8	283	1.3	297	0.87	626	1.2
Exports % GDP	280	30	316	41	297	39	893	37
ICT Compensation Share	280	0.041	316	0.045	297	0.048	893	0.045
Non-ICT Compensation Share	280	0.46	316	0.43	297	0.32	893	0.4
Labor Compensation Share	280	0.5	316	0.53	297	0.63	893	0.55

Source: The Conference Board (2014a) and World Bank (2014), own calculations.

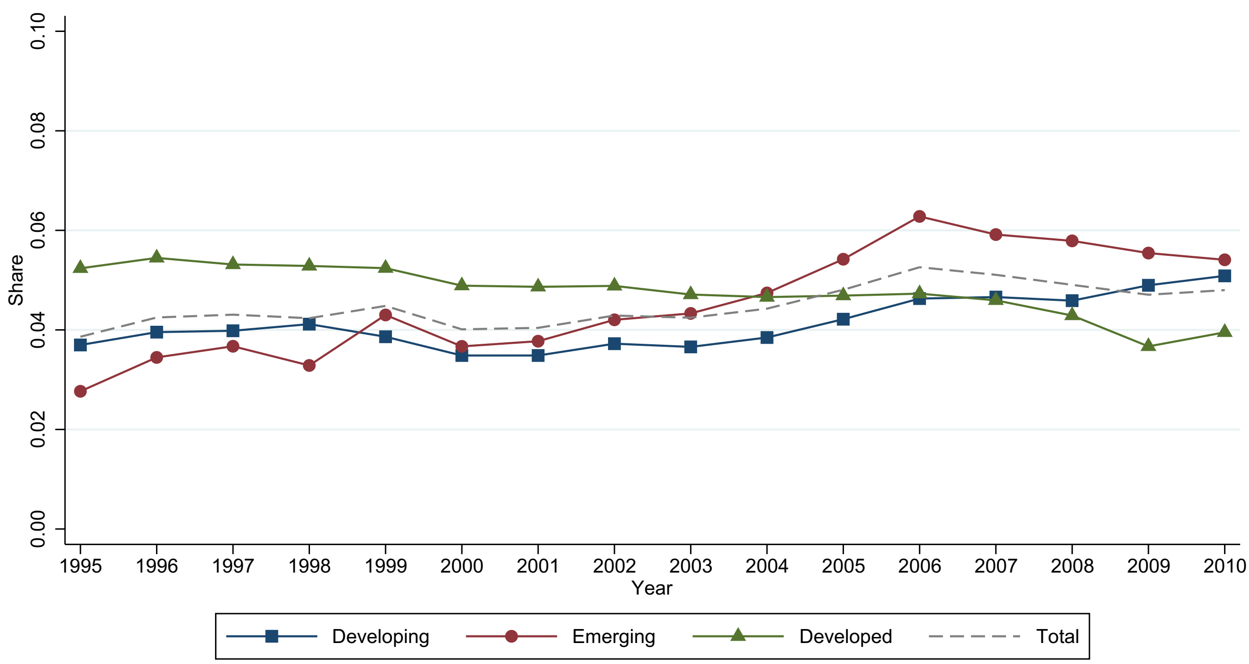

Fig. 3.1. Average ICT capital compensation share. *Source:* The Conference Board (2014a).

Another alternative would be the mean group (MG) estimator that allows for parameter heterogeneity between countries as described in Pesaran and Smith (1995). With this type of estimator, country specific estimations are carried out and later averaged across these countries. Given the still rather short time span in this empirical application, this estimator (and other heterogeneous parameter estimators) should be viable in future research with data sets cover a longer period of time.[29]

4. Empirical results

Table 4.1 presents the results for the Cobb-Douglas production function estimation of the full sample for the period 1995–2010. The growth rate of ICT capital services shows, with coefficients between 0.087 and 0.088, a nearly identical output elasticity for the pooled OLS (column 1), the random effects (column 2) and the fixed effects specification (column 3).[30] This estimated elasticity is larger than the ICT capital compensation share of 0.045%[31] and might be an indication of unmeasured complementarities or spillovers of ICT capital. The fourth column shows a fixed effects IV regression, where ICT, the main variable of interest, is instrumented by its lagged value. Lagged ICT capital services growth rates can be seen as a relevant instrument. In both IV specifications, the Angrist-Pischke F-test shows values clearly

[29] Eberhardt, Helmers, and Strauss (2013) provide an in-depth review and empirical application of heterogeneous parameter estimators.

[30] A Hausman style test, indicating whether the random or the fixed effects estimator is more appropriate, is not practically feasible in a setting with robust standard errors. A Wooldridge test for autocorrelation in panel data models cannot be rejected for the full sample and all three country subgroups.

[31] See Table A.6 in the Appendix.

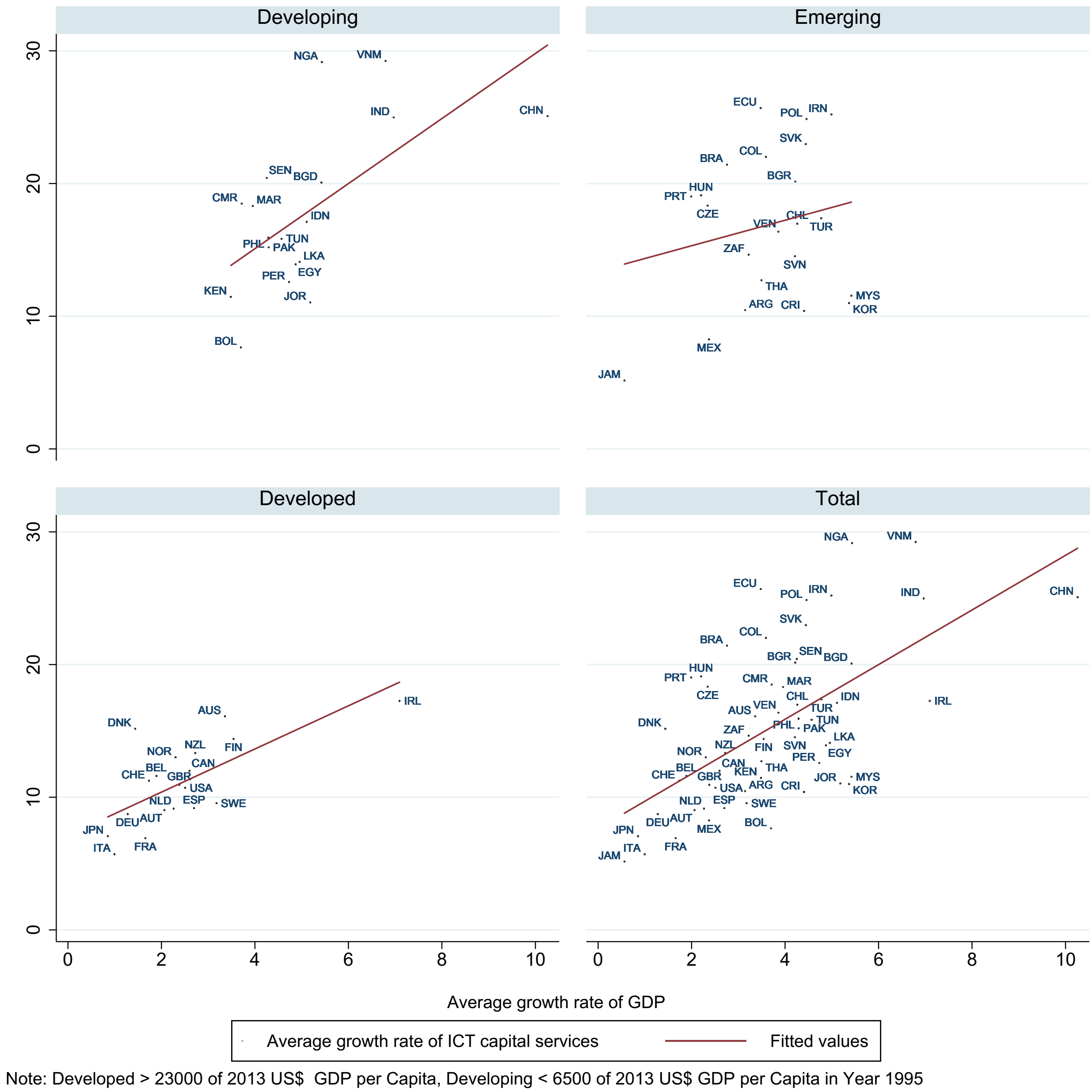

Fig. 3.2. GDP growth vs ICT capital services growth – average of 1995–2010. *Source:* The Conference Board (2014a).

Table 4.1
Dependent variable: $\Delta\ln(\mathrm{GDP})$ – 1995–2010 – full sample.

	(1) POLS	(2) RE	(3) FE	(4) IV
$\Delta\ln$(ICT Cap. Serv)	0.087^{***}	0.088^{***}	0.087^{***}	0.049^{***}
	(0.015)	(0.014)	(0.017)	(0.018)
$\Delta\ln$(N.ICT Cap. Serv.)	0.354^{***}	0.311^{***}	0.236^{***}	0.217^{***}
	(0.079)	(0.064)	(0.056)	(0.065)
$\Delta\ln$(Labor Serv.)	0.361^{***}	0.340^{***}	0.334^{***}	0.342^{***}
	(0.054)	(0.055)	(0.061)	(0.044)
Constant	0.994^{***}	1.147^{***}	1.412^{***}	
	(0.276)	(0.240)	(0.257)	
Year Dummies	Yes	Yes	Yes	Yes
Adjusted R^2	0.502	0.511	0.408	0.373
Observations	893	893	893	815

Clustered standard errors in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 4.2
Dependent variable: Δ ln(GDP) – 1995–2010 – full sample – augmented regression.

	(1) POLS	(2) RE	(3) FE	(4) IV
Δ ln(ICT Cap. Serv)	0.089***	0.092***	0.100***	0.066***
	(0.014)	(0.014)	(0.018)	(0.018)
Δ ln(N.ICT Cap. Serv.)	0.351***	0.312***	0.283***	0.256***
	(0.082)	(0.064)	(0.051)	(0.064)
Δ ln(Labor Serv.)	0.368***	0.341***	0.313***	0.323***
	(0.055)	(0.055)	(0.059)	(0.043)
Exports % GDP	0.012*	0.019***	0.069***	0.073***
	(0.006)	(0.006)	(0.015)	(0.015)
Constant	0.469	0.344	−1.642**	
	(0.348)	(0.307)	(0.672)	
Year Dummies	Yes	Yes	Yes	Yes
Adjusted R^2	0.508	0.516	0.426	0.398
Observations	893	893	893	815

Clustered standard errors in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 4.3
Dependent variable: Δ ln(GDP) – 1995–2010 – split sample.

	Developing			Emerging			Developed		
	(1) POLS	(2) RE	(3) FE	(4) POLS	(5) RE	(6) FE	(7) POLS	(8) RE	(9) FE
Δ ln(ICT Cap. Serv)	0.066***	0.074***	0.077***	0.059**	0.055**	0.048*	0.084**	0.077**	0.048**
	(0.015)	(0.014)	(0.022)	(0.028)	(0.024)	(0.028)	(0.037)	(0.033)	(0.023)
Δ ln(N.ICT Cap. Serv.)	0.296**	0.269**	0.177*	0.309***	0.281***	0.246***	0.212	0.161	−0.142
	(0.127)	(0.117)	(0.101)	(0.068)	(0.062)	(0.077)	(0.167)	(0.160)	(0.096)
Δ ln(Labor Serv.)	−0.092	−0.037	0.002	0.390***	0.402***	0.419***	0.446***	0.432***	0.397***
	(0.102)	(0.062)	(0.042)	(0.071)	(0.082)	(0.098)	(0.081)	(0.081)	(0.077)
Constant	2.715***	2.592***	2.919***	1.538**	1.630***	1.813***	1.198**	1.366***	1.977***
	(0.511)	(0.500)	(0.682)	(0.586)	(0.510)	(0.414)	(0.450)	(0.407)	(0.364)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.353	0.393	0.233	0.453	0.483	0.469	0.705	0.722	0.710
Observations	280	280	280	316	316	316	297	297	297

Clustered standard errors in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 4.4
Dependent variable: Δ ln(GDP) – 1995–2010 – split sample – augmented regression.

	Developing			Emerging			Developed		
	(1) POLS	(2) RE	(3) FE	(4) POLS	(5) RE	(6) FE	(7) POLS	(8) RE	(9) FE
Δ ln(ICT Cap. Serv)	0.067***	0.078***	0.103***	0.071**	0.066***	0.057**	0.073**	0.066**	0.049**
	(0.015)	(0.015)	(0.030)	(0.026)	(0.023)	(0.026)	(0.033)	(0.028)	(0.021)
Δ ln(N.ICT Cap. Serv.)	0.295**	0.260**	0.198*	0.301***	0.286***	0.310***	0.214	0.145	−0.093
	(0.128)	(0.119)	(0.099)	(0.065)	(0.058)	(0.070)	(0.138)	(0.120)	(0.093)
Δ ln(Labor Serv.)	−0.091	−0.032	−0.005	0.420***	0.416***	0.402***	0.416***	0.405***	0.361***
	(0.104)	(0.058)	(0.040)	(0.072)	(0.082)	(0.093)	(0.073)	(0.074)	(0.078)
Exports % GDP	0.002	0.013	0.049**	0.023***	0.026***	0.065***	0.014	0.018	0.063***
	(0.016)	(0.015)	(0.022)	(0.007)	(0.006)	(0.022)	(0.010)	(0.011)	(0.014)
Constant	2.652***	2.105**	0.705	0.199	0.245	−1.443	0.756	0.766	−0.654
	(0.909)	(0.859)	(1.111)	(0.604)	(0.533)	(1.140)	(0.612)	(0.614)	(0.755)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.351	0.389	0.241	0.482	0.512	0.480	0.718	0.733	0.722
Observations	280	280	280	316	316	316	297	297	297

Clustered standard errors in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

above the critical value of 10. Kleibergen-Paap LM-tests, which are heteroscedasticity-robust tests for underidentification, are always rejected, meaning that the model is identified. The ICT coefficient of 0.049 indicates that there might be an upward endogeneity bias in specifications (1)–(3) of Table 4.1. The estimated coefficients for labor services, which is a combined measure of labor quantity and composition, range from 0.334 to 0.361 depending on the type of estimator. The coefficients of non-ICT capital range from 0.217 to 0.354 with the smallest values for the fixed effects and IV estimator.

Table 4.2 shows the regression results for an augmented Cobb-Douglas production function. In comparison to Eq. (1), the additional variable is the share of exports in total GDP as an indicator for a country's openness to trade. This indicator controls for differences in production technology between countries. The time-varying share of exports in total GDP is at the same time a proxy for quite a lot of different country characteristics (regulation, infrastructure, knowledge inflows etc.). Time persistent, country specific factors are captured by the country fixed effects. Using additional control variables like the World Bank Worldwide

Governance Indicators or the Barro-Lee Educational Attainment Data, further robustness tests have been carried out. The results were qualitatively the same. One explanation for that could be that the Worldwide Governance Indicators in particular often don't vary much on a year-by-year basis. Therefore, the parsimonious model with just the share of exports in total GDP as additional control variable is used.

With the export coefficients being clearly positive and significant, openness to trade is correlated with higher GDP growth rates. The output elasticity of ICT capital services in this augmented production function setting is now larger than in Table 4.1 for all specifications. The largest differences occur in the FE and IV specification with ICT coefficients of 0.10 and 0.066. One explanation for this result could be that the export share is capturing otherwise unmeasured cross-country heterogeneity. The same applies to the output elasticity of non-ICT capital. The coefficients in the fixed effects and panel IV specification with the augmented Cobb-Douglas production function are also clearly larger than in Table 4.1 with the simple Cobb-Douglas production function.

Comparing the contribution of ICT to growth in developing, emerging and developed countries is the main goal of this paper. Therefore, in the next step, the production functions are estimated for each country subgroup individually. Table 4.3 presents results for the split-sample regressions of the three country subgroups for the period 1995–2010. The coefficients for ICT show rather modest differences between the country subgroups. The ICT output elasticity of the fixed effects specification shows the same coefficients (in both cases 0.048) for both the developed and emerging countries. The contribution of ICT is slightly larger in developing countries, which have a coefficient of 0.077. The estimated ICT coefficients are equal (for developed countries) or larger (for emerging and developing countries) than the ICT factor compensation shares shown in Tables A.3, A.4 and A.5. The additional productivity gains could be triggered by "ICT-related spillovers or network effects" as ICT may lower transaction costs and speed up the process of knowledge creation.

However, a test of the equality of the fixed effects ICT coefficients across the three country subgroups could not be rejected. A Chow-type test on the differences of the ICT coefficients between the country subgroups is conducted as follows: (1) estimate the OLS regression with country dummies, (2) estimate a seemingly unrelated regression using the Stata *suest* command and (3) test whether the difference between the coefficients of the country pairs is zero. All tests of the country pairs do no reject the differences between the ICT coefficients being zero. In other words, although the ICT coefficient of 0.077 is larger in developing than in developed and emerging countries, there is no statistically significant difference between the country subgroups. Hence, even with this slightly higher point estimate, there is no clear evidence to support the hypothesis of developing countries 'leapfrogging' through ICT as described in Steinmueller (2001). It should be noted that the coefficients for non-ICT capital services in developed countries and labor services in developing countries are insignificant. This could be an indicator for measurement error. The fact that labor services in developing countries are based mainly on the number of employees and not on the preferable measure of total hours worked could also be part of the explanation for the insignificant coefficients.

Table 4.4 shows the results for the country subgroups for the same robustness check as for the full sample specification in Table 4.2 before. The standard Cobb-Douglas production function for the three country subgroups is again augmented by the export share in GDP. The export share in emerging countries is, as in the full sample, positively correlated with GDP growth, whereas for developing and developed countries, only the FE specifications show significant effects. The fixed effects specifications (columns 3, 6 and 9) show slightly larger coefficients than before, ranging from 0.049 in developed countries to 0.103 in developing countries.

Tables A.7 and A.8 in the Appendix present results for a split into the subperiods 1995–2000 and 2000–2010 and reveal a lower ICT coefficient in the earlier period. These results seems to be largely driven by emerging countries between 1995 and 2000, which display insignificant coefficients for this subperiod.[32] However, during the subperiod 2000–2010, the ICT coefficients in emerging countries are slightly larger than in developing and developed countries. These substantial productivity gains of ICT capital in emerging countries cannot, however, provide statistically significant evidence for the 'leapfrogging' hypothesis.

5. Concluding remarks

This paper investigates the importance of ICT for economic growth based on a sample of 59 countries over the period 1995–2010. The regression of the full sample of countries reveals an output elasticity of ICT that is larger than the ICT factor compensation share, indicating possible spillovers and complementarities of investments in ICT. These excess returns confirm the positive link between ICT and economic development.

The main question, however, is whether the gains from investments in ICT are different between developing, emerging and developed countries. This question is answered, by running several independent regressions for the subsamples of developing, emerging and developed countries and subsequently testing the equality of estimated coefficients between the country subgroups. The regressions for the three country subsamples reveal rather small differences in the output elasticities of ICT between developing, emerging and developed countries. The test on the equality of these estimated coefficients could not be rejected, despite the coefficients being somewhat larger for the developing and emerging countries. So there is no clear statistical indication that developing and emerging countries are gaining more from investments in ICT than developed economies. Therefore, based on this analysis, the macroeconometric validity of the 'leapfrogging' through ICT argument as identified by Steinmueller (2001) remains rather questionable. Nevertheless, ICT still contributes substantially to economic growth not only in developed but also in developing and emerging countries.

Two additional issues are worth mentioning. While the present data set covers the majority of developed countries, emerging and developing countries are only represented to a limited extent. The list of developing and emerging countries with data on ICT capital input might not be randomly defined, but rather represents countries with larger GDP growth during the sample period. Therefore, a selection bias into the direction of countries that use ICT more efficiently might be present, resulting in the generalizability of the results being only valid to a certain extent. Furthermore, not only economic but also political and societal aspects such as the easier access to information should be taken into account when investigating the impact of ICT in developing and emerging countries.

Additional analyses based on larger sample sizes with respect to the time span as well as to the number of countries per subgroup should be able to use more refined econometric methods, helping to confirm the current results. This is especially important with regard to the potential endogeneity issues within macro-level production function estimations. Furthermore, complementary firm-level studies could help to gain deeper insights into the productivity effects of ICT in developing and emerging countries.

[32] See Tables A.9 and A.10.

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Appendix A

A.1. Additional graphs

Fig. A.1.

A.2. Additional tables

Tables A.1–A.10.

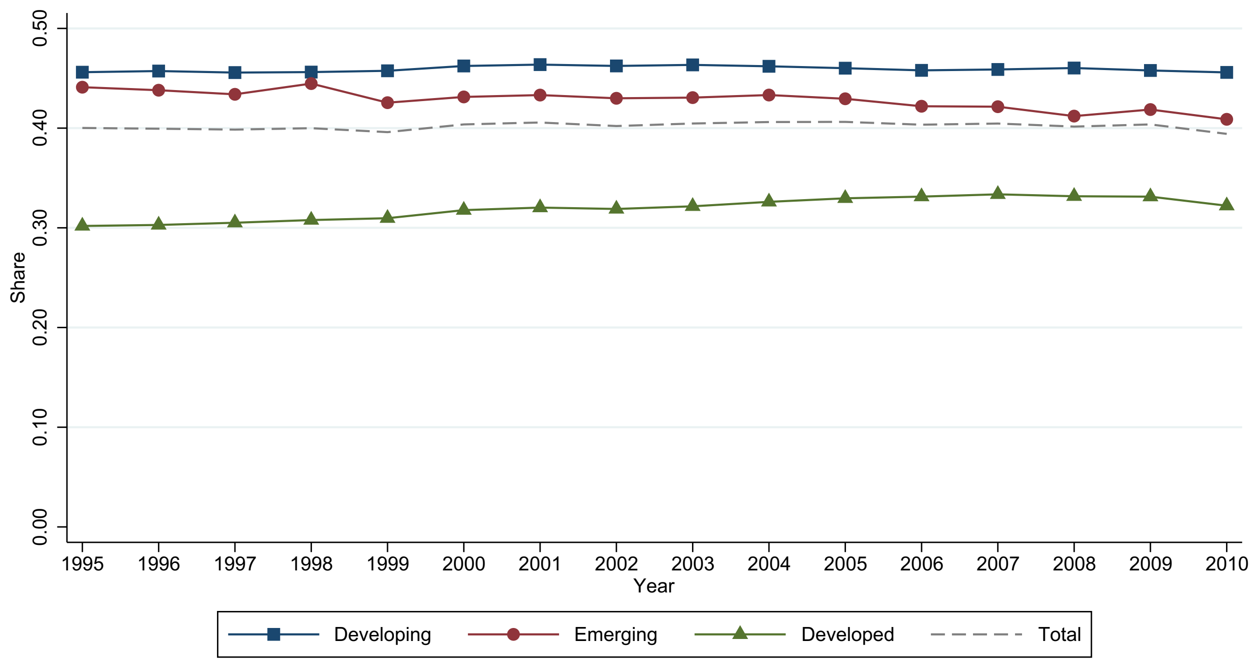

Fig. A.1. Average non-ICT capital compensation share. *Source:* The Conference Board (2014a).

Table A.1
GDP per capita in 2013 US$ in Year 2010.

Developing				Emerging				Developed			
1.	Peru	PER	9009	1.	South Korea	KOR	31,319	1.	Norway	NOR	57,706
2.	China	CHN	8910	2.	Slovenia	SVN	29,384	2.	United States	USA	50,605
3.	Tunisia	TUN	8173	3.	Czech Republic	CZE	27,287	3.	Switzerland	CHE	48,002
4.	Jordan	JOR	5623	4.	Portugal	PRT	23,535	4.	Australia	AUS	45,814
5.	Egypt	EGY	5516	5.	Slovak Republic	SVK	22,576	5.	Austria	AUT	44,021
6.	Indonesia	IDN	4921	6.	Poland	POL	19,629	6.	Netherlands	NLD	43,977
7.	Sri Lanka	LKA	4870	7.	Hungary	HUN	19,054	7.	Sweden	SWE	43,236
8.	Bolivia	BOL	4517	8.	Chile	CHL	15,284	8.	Canada	CAN	42,855
9.	Morocco	MAR	4089	9.	Malaysia	MYS	14,989	9.	Belgium	BEL	41,118
10.	Philippines	PHL	3817	10.	Mexico	MEX	14,185	10.	United Kingdom	GBR	39,545
11.	India	IND	3764	11.	Argentina	ARG	13,787	11.	Denmark	DNK	39,299
12.	Vietnam	VNM	3691	12.	Iran	IRN	13,448	12.	Germany	DEU	38,892
13.	Pakistan	PAK	2800	13.	Costa Rica	CRI	13,388	13.	Ireland	IRL	38,869
14.	Cameroon	CMR	2136	14.	Bulgaria	BGR	12,622	14.	Finland	FIN	38,007
15.	Nigeria	NGA	2098	15.	Turkey	TUR	12,241	15.	Japan	JPN	36,765
16.	Bangladesh	BGD	1762	16.	Venezuela	VEN	11,147	16.	France	FRA	35,606
17.	Senegal	SEN	1620	17.	Thailand	THA	10,220	17.	Italy	ITA	32,664
18.	Kenya	KEN	1442	18.	Brazil	BRA	9958	18.	New Zealand	NZL	31,949
				19.	Jamaica	JAM	9893	19.	Spain	ESP	31,669
				20.	Colombia	COL	9181				
				21.	South Africa	ZAF	9103				
				22.	Ecuador	ECU	8051				

Table A.2
List of dropped observations.

Country	Year	GDPg	LABg	ICTg	NICTg	ICT Comp.	Error	Scatter	DFBETA	Hadi
ARG	1995	−2.93	−10.02	7.66	2.92	0.044	0	0	0	1
ARG	1997	7.45	11.84	7.66	3.16	0.040	0	0	0	1
ARG	2002	−11.43	−13.09	1.58	−0.99	0.013	0	0	0	1
ARG	2003	7.60	16.43	0.38	−2.12	0.000	1	0	0	1
ARG	2004	7.04	14.27	4.94	1.09	0.049	0	0	0	1
ARG	2009	−3.28	−3.59	15.78	3.59	0.000	1	0	0	0
ARG	2010	8.00	5.48	13.59	3.65	0.000	1	0	0	0
BGR	1996	−9.46	3.09	14.07	2.11	0.048	0	0	0	1
BGR	2009	−5.63	−3.79	20.07	10.14	0.068	0	0	0	1
BRA	2010	6.64	4.44	15.52	3.72	0.000	1	0	0	0
CHN	1996	2.04	1.41	34.86	10.55	0.028	0	0	1	0
CHN	1998	0.29	1.56	28.49	9.23	0.036	0	0	1	0
COL	1999	−4.29	−1.56	32.35	1.52	0.019	0	1	0	0
CZE	2004	4.63	0.98	3.49	4.98	0.042	0	1	0	0
CZE	2005	6.53	2.59	2.63	5.15	0.037	0	1	0	0
CZE	2006	6.78	0.56	3.07	5.40	0.035	0	1	0	0
ECU	1999	−4.86	−2.04	28.58	0.51	0.011	0	1	0	0
FIN	1995	3.89	2.52	24.28	−2.51	0.029	0	1	0	0
FIN	1996	3.51	1.78	22.65	−0.96	0.035	0	1	0	0
FIN	2009	−8.93	−3.30	11.36	1.40	0.079	0	0	1	0
IDN	1998	−14.08	1.07	3.71	6.86	0.021	0	1	0	1
IRL	2008	−2.18	−1.47	14.60	5.56	0.050	0	0	1	0
IRL	2009	−6.60	−9.63	11.94	4.33	0.045	0	0	1	1
IRL	2010	−1.07	−4.47	10.36	2.73	0.042	0	0	1	0
KOR	1998	−5.87	−7.38	12.73	4.98	0.057	0	0	0	1
LKA	2009	3.48	0.30	31.47	3.89	0.189	1	0	0	0
LKA	2010	7.71	3.33	29.66	3.70	0.341	1	0	0	0
MAR	1995	−6.81	4.08	10.62	0.55	0.046	0	0	1	0
MAR	1996	11.53	3.06	11.24	0.83	0.047	0	0	1	0
MYS	1995	9.38	5.82	11.76	13.92	0.128	0	0	0	1
MYS	1998	−7.64	0.59	10.65	6.67	0.123	0	0	0	1
NGA	2002	19.35	3.48	30.79	2.09	0.021	0	1	0	1
POL	1999	4.42	−4.36	40.88	4.84	0.017	0	0	0	1
SVN	2009	−8.28	−7.67	11.96	3.41	0.052	0	0	0	1
SWE	2009	−5.16	−2.80	8.60	2.00	0.056	0	0	1	0
TUR	2008	0.66	1.82	21.09	6.55	0.000	1	0	0	0
TUR	2009	−4.95	−0.39	16.05	4.93	0.000	1	0	0	0
TUR	2010	8.62	6.02	13.35	4.80	0.000	1	0	0	0
VEN	2002	−9.27	0.71	11.86	0.25	0.023	0	1	0	1
VEN	2003	−8.07	0.99	8.54	−1.95	0.035	0	1	0	0
VEN	2004	16.79	6.88	11.06	−2.33	0.052	0	0	0	1
VEN	2008	5.14	4.57	24.99	4.80	0.002	1	0	0	0
VEN	2009	−3.25	2.74	20.12	4.05	0.002	1	0	0	0
VEN	2010	−1.50	2.71	15.37	3.39	0.002	1	0	0	0

Source: The Conference Board (2014a).

Table A.3
Summary statistics: developing countries – 1995–2010.

	N	Mean	Median	SD	Min	Max
GDP per Capita	280	3325	3085	1757	998	9009
Δ ln(GDP)	280	5.1	5	2.5	−2.3	14
Δ ln(ICT Cap. Serv)	280	18	16	8.1	−0.46	44
Δ ln(N.ICT Cap. Serv.)	280	4.8	3.7	3	−0.56	13
Δ ln(Labor Serv.)	280	2.8	2.7	1.8	−2.9	10
Δ ln(Labor Composition)	280	0.25	0.28	0.16	−0.12	0.73
Δ ln(Labor Quantity)	280	2.6	2.4	1.9	−3.2	10
Δ ln(Employees)	280	2.5	2.4	1.8	−5	9.5
Δ ln(Hours Worked)	46	2.8	2.2	2.5	−3.2	10
Exports % GDP	280	30	29	13	10	72
ICT Compensation Share	280	0.041	0.037	0.021	0.009	0.11
Non-ICT Compensation Share	280	0.46	0.46	0.024	0.39	0.54
Labor Compensation Share	280	0.5	0.5	0.015	0.42	0.57

Source: The Conference Board (2014a) and World Bank (2014), own calculations.

Table A.4
Summary statistics: emerging countries – 1995–2010.

	N	Mean	Median	SD	Min	Max
GDP per Capita	316	13,367	11,486	5915	6433	31,451
Δ ln(GDP)	316	3.6	3.9	3	−8	10
Δ ln(ICT Cap. Serv)	316	17	17	7.3	1	40
Δ ln(N.ICT Cap. Serv.)	316	3.9	3.6	2.6	−1.2	13
Δ ln(Labor Serv.)	316	1.9	1.9	2.5	−4.1	10
Δ ln(Labor Composition)	316	0.48	0.43	0.52	−2.8	4.4
Δ ln(Labor Quantity)	316	1.4	1.4	2.5	−4.6	9.8
Δ ln(Employees)	316	1.5	1.6	2.2	−4.6	9.5
Δ ln(Hours Worked)	283	1.3	1.3	2.5	−4.6	9.8
Exports % GDP	316	41	35	23	6.6	121
ICT Compensation Share	316	0.045	0.044	0.024	0.0014	0.12
Non-ICT Compensation Share	316	0.43	0.45	0.1	0.14	0.66
Labor Compensation Share	316	0.53	0.5	0.097	0.32	0.84

Source: The Conference Board (2014a) and World Bank (2014), own calculations.

Table A.5
Summary statistics: developed countries – 1995–2010.

	N	Mean	Median	SD	Min	Max
GDP per Capita	297	38,044	37,098	6575	23,913	58,957
Δ ln(GDP)	297	2.4	2.6	2.3	−5.8	11
Δ ln(ICT Cap. Serv)	297	11	11	4.5	0.35	23
Δ ln(N.ICT Cap. Serv.)	297	2.3	2.2	1.4	−0.45	8.1
Δ ln(Labor Serv.)	297	1.3	1.4	1.8	−5.7	6.5
Δ ln(Labor Composition)	297	0.4	0.33	0.35	−1.2	1.6
Δ ln(Labor Quantity)	297	0.87	0.99	1.8	−6.3	5.7
Δ ln(Employees)	297	1.2	1.2	1.6	−6.7	8.1
Δ ln(Hours Worked)	297	0.87	0.99	1.8	−6.3	5.7
Exports % GDP	297	39	38	20	9.1	99.7
ICT Compensation Share	297	0.048	0.047	0.011	0.023	0.083
Non-ICT Compensation Share	297	0.32	0.31	0.058	0.2	0.52
Labor Compensation Share	297	0.63	0.64	0.056	0.44	0.75

Source: The Conference Board (2014a) and World Bank (2014), own calculations.

Table A.6
Summary statistics: 1995–2010 – full sample.

	N	Mean	Median	SD	Min	Max
GDP per Capita	893	18,426	11,818	15,377	998	58,957
Δ ln(GDP)	893	3.7	3.8	2.8	−8	14
Δ ln(ICT Cap. Serv)	893	15	14	7.4	−0.46	44
Δ ln(N.ICT Cap. Serv.)	893	3.6	2.9	2.6	−1.2	13
Δ ln(Labor Serv.)	893	2	2	2.2	−5.7	10
Δ ln(Labor Composition)	893	0.38	0.32	0.39	−2.8	4.4
Δ ln(Labor Quantity)	893	1.6	1.6	2.2	−6.3	10
Δ ln(Employees)	893	1.7	1.7	1.9	−6.7	9.5
Δ ln(Hours Worked)	626	1.2	1.2	2.2	−6.3	10
Exports % GDP	893	37	32	20	6.6	121
ICT Compensation Share	893	0.045	0.044	0.02	0.0014	0.12
Non-ICT Compensation Share	893	0.4	0.43	0.092	0.14	0.66
Labor Compensation Share	893	0.55	0.5	0.088	0.32	0.84

Source: The Conference Board (2014a) and World Bank (2014), own calculations.

Table A.7
Dependent variable: Δ ln(GDP) – 1995–2000 – full sample.

	(1) POLS	(2) RE	(3) FE
Δ ln(ICT Cap. Serv)	0.063**	0.065***	0.072*
	(0.024)	(0.024)	(0.043)
Δ ln(N.ICT Cap. Serv.)	0.334***	0.317***	0.218**
	(0.059)	(0.058)	(0.090)
Δ ln(Labor Serv.)	0.402***	0.380***	0.317***
	(0.085)	(0.087)	(0.107)
Exports % GDP	0.019*	0.020**	0.052
	(0.010)	(0.010)	(0.036)
Constant	0.538	0.539	−0.365
	(0.609)	(0.584)	(1.597)
Year Dummies	Yes	Yes	Yes
Adjusted R^2	0.328	0.345	0.139
Observations	335	335	335

Clustered standard errors in parentheses, *p < .10, **p < .05, ***p < .01.

Table A.8
Dependent variable: Δ ln(GDP) – 2000–2010 – full sample.

	(1) POLS	(2) RE	(3) FE
Δ ln(ICT Cap. Serv)	0.105***	0.096***	0.099***
	(0.015)	(0.015)	(0.020)
Δ ln(N.ICT Cap. Serv.)	0.363***	0.352***	0.342***
	(0.110)	(0.072)	(0.066)
Δ ln(Labor Serv.)	0.312***	0.275***	0.252***
	(0.063)	(0.050)	(0.052)
Exports % GDP	0.011*	0.021***	0.078***
	(0.006)	(0.007)	(0.020)
Constant	0.332	0.156	−2.107**
	(0.357)	(0.327)	(0.856)
Year Dummies	Yes	Yes	Yes
Adjusted R^2	0.590	0.592	0.539
Observations	616	616	616

Clustered standard errors in parentheses, *p < .10, **p < .05, ***p < 0.01.

Table A.9
Dependent variable: Δ ln(GDP) – 1995–2000.

	Developing			Emerging			Developed		
	(1) POLS	(2) RE	(3) FE	(4) POLS	(5) RE	(6) FE	(7) POLS	(8) RE	(9) FE
Δ ln(ICT Cap. Serv)	0.093***	0.098***	0.198***	0.060	0.058	−0.015	0.144**	0.104**	0.061*
	(0.030)	(0.029)	(0.057)	(0.040)	(0.041)	(0.049)	(0.062)	(0.047)	(0.032)
Δ ln(N.ICT Cap. Serv.)	0.230**	0.227**	0.127	0.314***	0.300***	0.230	0.488**	0.269*	−0.010
	(0.092)	(0.091)	(0.144)	(0.073)	(0.072)	(0.134)	(0.190)	(0.139)	(0.175)
Δ ln(Labor Serv.)	0.005	0.006	0.005	0.531***	0.535***	0.573**	0.338***	0.318***	0.265**
	(0.052)	(0.047)	(0.054)	(0.144)	(0.148)	(0.205)	(0.096)	(0.097)	(0.113)
Exports % GDP	−0.017	−0.017	0.033	0.027***	0.028***	0.082	0.025	0.033*	0.075
	(0.013)	(0.013)	(0.055)	(0.009)	(0.009)	(0.062)	(0.015)	(0.018)	(0.056)
Constant	2.462***	2.403***	−0.278	−0.104	−0.091	−0.871	−1.004	−0.130	−0.340
	(0.664)	(0.676)	(2.215)	(1.124)	(1.145)	(2.566)	(1.052)	(0.864)	(2.263)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.347	0.404	0.170	0.265	0.320	0.199	0.642	0.653	0.290
Observations	103	103	103	120	120	120	112	112	112

Clustered standard errors in parentheses, *p < .10, **p < .05, ***p < .01.

Table A.10
Dependent variable: Δ ln(GDP) – 2000–2010.

	Developing			Emerging			Developed		
	(1) POLS	(2) RE	(3) FE	(4) POLS	(5) RE	(6) FE	(7) POLS	(8) RE	(9) FE
Δ ln(ICT Cap. Serv)	0.063*	0.062***	0.065	0.088***	0.084***	0.069*	0.059*	0.059**	0.064**
	(0.033)	(0.023)	(0.038)	(0.026)	(0.023)	(0.039)	(0.030)	(0.029)	(0.028)
Δ ln(N.ICT Cap. Serv.)	0.311	0.240*	0.158	0.303***	0.341***	0.459***	0.155	0.153	−0.004
	(0.182)	(0.130)	(0.118)	(0.095)	(0.090)	(0.087)	(0.122)	(0.121)	(0.124)
Δ ln(Labor Serv.)	−0.139	−0.018	0.013	0.346***	0.315***	0.269***	0.370***	0.363***	0.301***
	(0.153)	(0.072)	(0.062)	(0.066)	(0.068)	(0.075)	(0.063)	(0.066)	(0.062)
Exports % GDP	0.012	0.032*	0.052*	0.023***	0.023***	0.056	0.007	0.008	0.104***
	(0.022)	(0.019)	(0.025)	(0.007)	(0.008)	(0.035)	(0.007)	(0.007)	(0.022)
Constant	2.427**	1.901*	1.568	0.065	0.115	−1.431	1.200**	1.179**	−2.561**
	(1.121)	(1.143)	(1.851)	(0.688)	(0.576)	(1.829)	(0.480)	(0.481)	(1.119)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.340	0.367	0.261	0.619	0.642	0.630	0.757	0.774	0.780
Observations	195	195	195	217	217	217	204	204	204

Clustered standard errors in parentheses, $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

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